

DiffAudit: Evidence-Contracted Membership Leakage Auditing for Diffusion Models

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Abstract—Membership inference against diffusion models is often reported as a scalar attack score. Security/privacy audit consumption asks a different measurement question: when does that score become reusable evidence for a specific target, split, observable, metric, and consumer claim? We present DiffAudit, a claim-admission protocol that treats the unit of analysis as a row-bound evidence packet rather than a method name or headline AUC. A packet is admitted only when the target, split, row-level observable, metric provenance, finite-tail basis, and consumer boundary are fixed; portability claims also require a non-adjacent surface. Applied to the current evidence bank, DiffAudit admits five heterogeneous contract rows and isolates a high-AUC repeated-response H2 output-cloud signal as a controlled candidate rather than an admitted result, because img2img portability and runtime-consumer gates fail. Selected artifact and search records are used only to audit claim promotion, not to estimate field-wide prevalence. DiffAudit therefore reframes diffusion MIA as claim-bound audit consumption: leakage claims are retained where the contract supports them, while otherwise promising signals remain candidate, support-only, or negative instead of becoming downstream audit claims.

I. INTRODUCTION

Diffusion models can memorize or overfit training examples in ways that enable membership inference attacks. The literature has produced strong black-box, gray-box, and white-box signals, including reconstruction similarity, posterior-error statistics, proximal initialization, conditional likelihood discrepancy, reconstruction-based black-box attacks, and gradient-based attacks [1], [2], [3], [4], [5]. However, paper-level scores are not automatically audit evidence. A model owner, regulator, or downstream runtime needs to know which target checkpoint was audited, what row IDs define members and nonmembers, whether scores exist per sample, how low-FPR tails should be interpreted, and which claims are blocked by missing artifacts.

DiffAudit is built around this distinction. We treat the measured object as an evidence packet, not as a method name or a single score. The contract turns artifact details into scientific controls: target identity, split semantics, score or response coverage, metric provenance, consumer boundary, and surface delta expose source-label confounding, prompt-only shortcuts, finite-tail overclaiming, unsupported consumer promotion, and non-replayable headline metrics. The measurement result is the gate outcome itself: which claim survives, which claim is blocked, and which missing surface caused the boundary.

The audit decision problem has three consumers: model owners deciding whether a checkpoint creates measurable risk, auditors replaying row-bound metrics, and downstream reports that must not promote candidates into user-facing claims.

DiffAudit is therefore not an attack-SOTA paper: it studies when a diffusion MIA result is eligible to support a stated audit claim.

We organize the case study around three measurement questions. RQ1 asks what minimum contract is needed before a diffusion MIA score can support a downstream audit claim. RQ2 asks which existing score or response packets become admissible under that contract. RQ3 asks what strong but non-admitted signals reveal about portability, source confounding, and finite-tail overclaiming. The answer is not a new attack leaderboard: five rows become admitted under bounded consumer language, while H2 output-cloud geometry, Tracing Roots feature packets, CommonCanvas response packets, ReDiffuse scouts, MIDST, and collaborator Stable Diffusion packets each support narrower candidate, support-only, or negative claims.

The paper therefore makes the following contributions:

- We formulate audit consumption, rather than attack accuracy alone, as the measurement problem for diffusion MIA, and define a six-gate evidence contract for claim-bound evidence packets.
- We instantiate the contract on five contract-admitted rows with shared metric fields (AUC, ASR, TPR@1%FPR, and TPR@0.1%FPR), finite-tail bases, provenance, quality/cost metadata, and consumer-boundary language that separates primary risk evidence from comparator and bridge rows.
- We include repeated-response output-cloud geometry as a contract stress test: the signal is strong under bounded controls, but remains non-admitted because the current evidence does not establish a portable or consumer-ready black-box audit row.
- We show how bounded negative and support evidence, plus selected claim-control records, block overbroad portability and artifact-availability claims without becoming method refutations or field-wide prevalence estimates.

II. RELATED WORK

A. Membership Inference and Evidence

Membership inference was introduced for discriminative machine-learning services as a way to infer whether a record was used during training [6]. Subsequent work connected the risk to overfitting and clarified the statistical interpretation of membership signals [7], [8]. Recent position work further argues that membership inference should not be treated as standalone proof that a model was trained on a particular

datum [9]. DiffAudit adopts this caution as an audit-design principle: a high score is useful only after its target, split, metric, and consumer boundary are explicit.

B. Diffusion Memorization and MIA

Diffusion models build on denoising training and latent generation pipelines [10], [11]. They expose privacy risk through both memorized generations and membership signals. Extraction and replication studies show that training examples or near-duplicates can appear in outputs [12], [13]. Diffusion MIA is already split by access surface. Gray-box posterior, trajectory, and conditional-likelihood methods expose intermediate statistics or likelihood surrogates [1], [2], [3], [14]. Black-box response and reconstruction methods compare queries with returned or regenerated outputs [15], [4]. White-box gradient attacks require stronger evaluator access [5]. Defended and realistic benchmark settings add further boundaries: DPDM changes the target through differential privacy [16], CopyMark and MIDST show that image and tabular benchmark contracts affect what an MIA result supports [17], [18], and dataset-level owner auditing such as CDI aggregates evidence across many images rather than proving an individual row [19]. DiffAudit does not rank these attacks; it asks when their artifacts expose enough target, split, row, metric, and consumer context to become reusable audit evidence.

C. Reporting and Reproducibility Artifacts

DiffAudit is closest to reporting and reproducibility work in machine learning, but it measures a different object. Reproducibility checklists emphasize data, code, hyperparameters, and evaluation conditions [20]; datasheets and model cards document dataset and model origins, uses, and limitations [21], [22]. Privacy audit admission additionally needs a threat model, row-level membership labels or responses, metric provenance, finite-tail denominators, and an allowed consumer claim. DiffAudit therefore adapts reporting artifacts into a measurement-validity test: an incomplete artifact is not automatically a failed reproduction, but it cannot support a stronger audit claim than its packet exposes.

III. AUDIT SURFACES

This paper studies an audit-admission problem, not a universal membership proof. The membership game is binary: for a fixed target and frozen member/nonmember split, an evidence packet scores whether each labeled sample row belonged to the target’s training set under a stated access mode. The audit output is the strongest claim that this packet can support. Query budgets, sample counts, and auxiliary access are packet-specific and are not normalized into a universal leaderboard. We explicitly exclude individual training-data proof, field-wide artifact prevalence, and exposing candidate rows as consumer-facing audit-report claims.

Diffusion MIA results are not interchangeable because they observe different surfaces. A black-box response packet can support an output-consumer claim if target identity and query

TABLE I
AUDIT SURFACES AND ALLOWED EVIDENCE ROLES.

Surface	Allowed role in DiffAudit
Black-box responses	Admitted only with fixed target, split, response coverage, and metric replay.
Gray-box trajectory/statistics	Admitted or candidate depending on provenance and consumer semantics.
White-box gradients	Upper-bound comparator unless the audit consumer has white-box access.
Feature packets	Support-only unless raw target/sample provenance and regeneration assets are available.
Collaborator/local packets	Stress tests unless public row binding and source-confounding checks pass.

rows are fixed. A gray-box trajectory or posterior-statistics packet can support a mechanism or evaluator claim, but requires a different consumer boundary. A white-box gradient packet is useful as an upper-bound risk comparator, not as a black-box report. Feature packets and collaborator-local scores can be scientifically useful while remaining support-only. Table I records the surfaces used in this paper. Access mode is not interchangeable: a row admitted under one surface is not automatically admitted under another.

IV. EVIDENCE CONTRACT

DiffAudit uses six gates before a result can support a portability or runtime audit claim. The target identity gate requires a fixed model architecture, checkpoint identity, training provenance, and accessibility. The split contract requires precise member and nonmember manifests. The score or response gate requires row-level coverage and labels. The metric gate requires AUC, ASR, TPR@1%FPR, TPR@0.1%FPR, nonmember denominator, and traceability to a score array or response packet. The consumer-boundary gate requires a stated consumer and allowed claim language. The surface-delta gate checks whether a claimed second asset actually differs in model, data, modality, or response contract rather than repeating the same family.

The gates are intentionally stricter than ordinary artifact availability. A repository can contain code without a fixed target checkpoint, a checkpoint can exist without immutable member/nonmember manifests, a score file can be scoreable without proving the same consumer boundary, and a strong candidate can still be unsafe to expose as a runtime audit row. DiffAudit therefore asks what claim an artifact supports, not whether an artifact is globally useful or whether the original paper is reproducible in every detail.

Low-FPR metrics are reported as finite empirical packet readouts. For example, TPR@0.1%FPR on 100 nonmembers has a false-positive budget of 0.1 and should not be described as a calibrated continuous sub-percent operating point. This rule is important because many diffusion MIA packets are small, and strict-tail claims can otherwise sound stronger than the evidence permits.

The consumer-boundary gate is the most conservative part of the contract. It records who may consume the result and what language must accompany it. A white-box gradient attack

TABLE II
DIFFAUDIT EVIDENCE-CONTRACT GATES.

Gate	Question answered
Target identity	Which model/checkpoint is being audited?
Split semantics	Which rows are members and nonmembers?
Score/response coverage	Is there row-level evidence for every sample?
Metric provenance	Can headline metrics be traced to rows or frozen responses?
Consumer boundary	Who may consume the result and as what claim?
Surface delta	Is a claimed second asset actually non-adjacent?

may be a valid upper-bound comparator but not a black-box product result; a feature-packet replay may be useful evidence for temporal trajectory leakage but not a raw image audit; a cross-source Stable Diffusion packet may be a stress test but not same-distribution membership evidence. These distinctions are enforced before writing the paper, not added as defensive caveats after a claim is made.

Gate labels are claim-relative adjudications. *Pass* means the packet is frozen, row-bound, and sufficient for the stated consumer claim; *Partial* means a useful surface exists but cannot support a stronger claim without more identity, provenance, or consumer binding; *Fail* means a missing or contradictory surface blocks that claim; and *N/A* means the gate is outside the claim type. Each label records the missing surface that would change the state, so the protocol produces an audit decision rather than a repository quality score.

V. MEASUREMENT PROTOCOL

DiffAudit stores each result as one of five states. An admitted row can be reported to downstream consumers under its stated boundary. A candidate row is scientifically useful but not consumer-ready. A support-only row can motivate or contextualize a mechanism but cannot serve as a direct benchmark. A watch item is tracked for future artifacts. A negative row answers a bounded decision question and usually closes a low-value expansion.

The protocol is applied claim-first. The reviewer first writes the intended claim c , then assembles an evidence packet $p = (T, S, O, M, C, \Delta)$ binding target T , split S , row-level observable O , metric provenance M , consumer boundary C , and optional surface delta Δ . DiffAudit then records $G(p, c) \in \{\text{Pass}, \text{Partial}, \text{Fail}, \text{N/A}\}^6$ in the fixed gate order of Table II. A gate can pass only when the source map exposes the required surface for the proposed claim; otherwise the label records the concrete missing or contradictory surface. The pair (p, c) is contract-admitted only when target, split, evidence, metric, and consumer gates pass and the delta gate either passes or is not applicable. If not, the first non-passing required gate selects the narrower evidence state: candidate for open portability/consumer gates, support-only for different access or feature surfaces, watch for artifact discovery without rows, and negative for a bounded route-selection answer.

TABLE III
EVIDENCE STATES USED BY DIFFAUDIT.

State	Meaning
Admitted	All gates pass for the stated consumer boundary; row may appear in audit reports.
Candidate	Scientifically useful signal, but at least one portability or consumer gate remains open.
Support-only	Useful context or mechanism evidence, but not a direct benchmark row.
Watch	Public signal worth tracking, with no row-bound evidence yet.
Negative	Bounded route-selection result that closes or narrows a hypothesis.

Metrics are recomputed from row-level scores or frozen response packets when available. We report AUC, attack success rate (ASR), $\text{TPR}@1\%\text{FPR}$, and $\text{TPR}@0.1\%\text{FPR}$, but we do not treat all four values equally. Low-FPR values are interpreted through the nonmember denominator: on a packet with 25 nonmembers, $\text{TPR}@0.1\%\text{FPR}$ has no continuous sub-percent meaning. This is why DiffAudit records both metrics and packet boundaries instead of publishing a leaderboard.

A. Evidence States

The evidence state is part of the measurement result. It is not a project management label. The same numerical score can be admitted in one setting, candidate-only in another, and support-only in a third if its target identity, split, row coverage, or consumer boundary changes.

B. Metric Replay

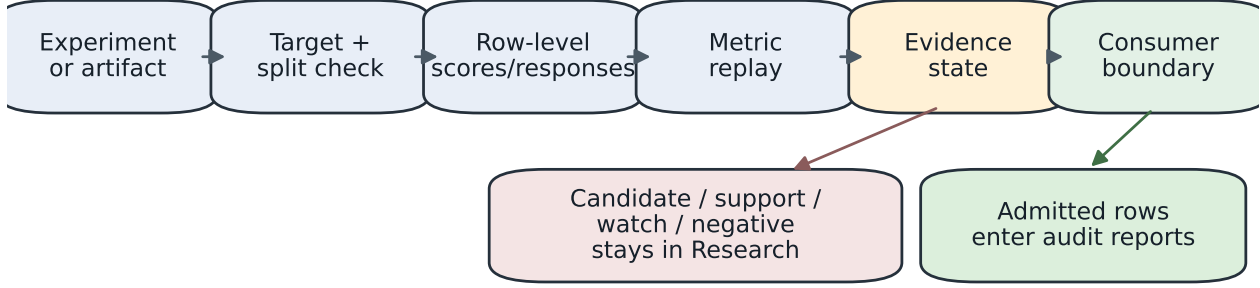
DiffAudit treats metric replay as a provenance check. When a score array exists, metrics are recomputed from row labels rather than copied from prose. When only responses exist, the scorer and response manifest must be fixed before the metric can be used. For a nonmember packet of size n_0 , FPR takes values on a discrete grid of $1/n_0$. Thus $\text{TPR}@0.1\%\text{FPR}$ is a strict empirical tail whenever $0.001n_0 < 1$. We still report the readout because it is useful for comparing packets, but the text must retain the denominator and avoid calibrated-risk language [8].

Table IV is used later as a claim-admission ablation: rows that look usable under ordinary code-or-metadata availability are blocked by different missing gates once row binding, metric replay, consumer boundary, and surface delta are evaluated.

VI. SELECTION BOUNDARY AND CLAIM-CONTROL EVIDENCE

This section records selection transparency, not a literature survey. The current claim-control set is intentionally small but heterogeneous: five admitted rows, one strong H2 candidate, bounded negative routes for ReDiffuse STL-10, CommonCanvas, and MIDST, positive feature-packet evidence from Tracing the Roots, and a source-confounding stress test from collaborator Stable Diffusion ReDiffuse. The concrete blockers are different: no downstream H2 admission or independent response asset, no distance-distinct img2img

DiffAudit evidence-contract pipeline



A metric is consumed only after provenance, finite-tail semantics, and boundary language are fixed.

Fig. 1. DiffAudit evidence-contract pipeline. The pipeline is intentionally claim-oriented: the same metric can become admitted evidence, a candidate case study, support-only context, or a bounded negative depending on provenance and consumer semantics.

TABLE IV

CANDIDATE AND SUPPORT EVIDENCE GATE MATRIX. LABELS ARE CLAIM-RELATIVE: A ROLE-PASS IS SUFFICIENT FOR THE STATED PAPER ROLE, NOT AUTOMATIC PRODUCT ADMISSION. ‘MIXED REPLAY’ MEANS ROW-SCORE REPLAY WHERE AVAILABLE AND SOURCE-DOCUMENTED POINT ESTIMATES OTHERWISE.

Route	Target	Split	Evidence	Metric	Boundary	Delta	Paper use
Admitted bundle (mixed replay)	Pass	Pass	Pass	Pass	Pass	N/A	Main audit rows
H2 output-cloud	Pass	Pass	Pass	Pass	Partial	Partial	Candidate mechanism
H2 img2img portability	Pass	Pass	Pass	Pass	Fail	Pass	Portability limit
ReDiffuse STL-10 scout	Pass	Pass	Pass	Pass	Fail	Pass	Bounded negative
CommonCanvas response packet	Partial	Pass	Pass	Pass	Fail	Pass	Bounded negative
Tracing Roots feature packet	Partial	Pass	Pass	Pass	Partial	Pass	Support-only
SD ReDiffuse collaborator	Partial	Partial	Pass	Pass	Fail	Pass	Source-confounded support

portability, weak CommonCanvas scorers under a partial target boundary, feature-packet rather than raw-image Tracing Roots evidence, and source-confounded collaborator labels. These rows exercise the evidence contract across admitted, candidate, support-only, and negative states, but they do not estimate artifact prevalence in the field.

Within this case study, protocol validity is evaluated as claim-promotion control, not as independent inter-rater reliability. If a row lacks target identity, row-level evidence, metric replay, or a consumer boundary, the gate vector should block admission even when the headline signal is strong. The counts below are diagnostic examples under a declared set, not evidence that public diffusion MIA artifacts generally fail.

A. Selection Trace

The current set was built from DiffAudit’s audited evidence surface. A standalone artifact-claim paper would require a larger frozen corpus or a second independent label review before making aggregate claims. We also instantiated a frozen supplementary metadata search on 2026-05-26, covering seven GitHub repository queries and five arXiv API metadata queries, including one noisy title-only query retained for auditability. The search was coded before aggregate claims and used no downloads or GPU runs. Its outcome is intentionally

modest: metadata discovery was easy, but no new row-bound public target/split/score or response packet appeared.

The selection trace makes the claim boundary inspectable rather than enlarging the denominator opportunistically. Fixed-search rows record the query, observed surface, no-download policy, inclusion/exclusion reason, gate labels, reviewer, and date. Empty-result queries constrain the search process, but they remain metadata-only and are never pooled with replayed score packets. The frozen batch records three re-found surfaces, one new code-public no-packet surface, eight relevant metadata-only rows, four empty fixed-query rows, one noisy query row, and zero new admitted audit rows.

Figure 2 makes the selection boundary visible. We separate claim-support levels before reading counts: L0 metadata discovery, L1 artifact inspection, L2 scoreable or replayed packets, and L3 admitted audit evidence. The fixed-search batch is metadata-only and is not pooled with replay rows as a reproduction denominator. A selected-corpus consistency pass found no invalid gate labels, no evidence/metric contradictions, and no label promotions; it only demoted four empty-result `delta_gate` labels from Partial to Fail. A standalone artifact-claim paper would still need a larger frozen corpus or a second independent label review before making aggregate claims.

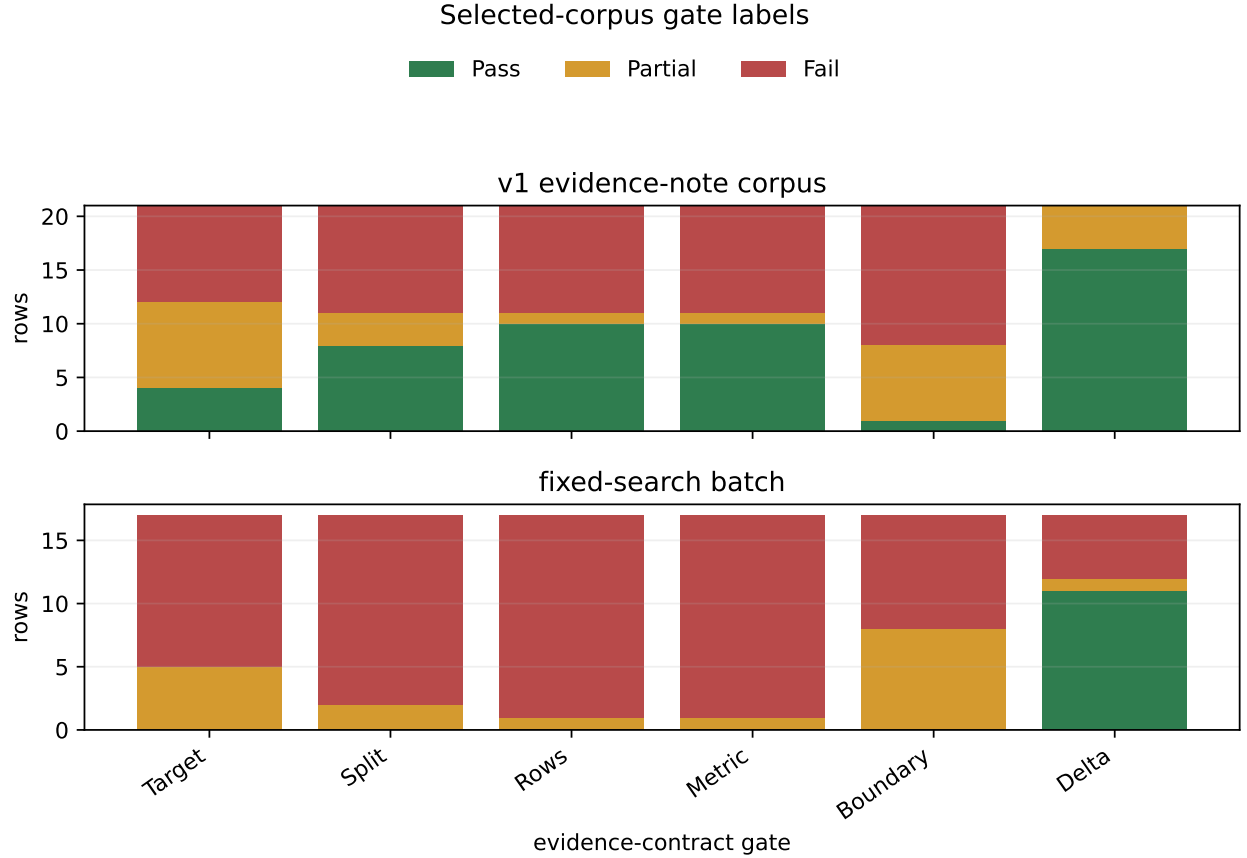


Fig. 2. Gate-label counts for the selected v1 evidence-note set and the frozen 2026-05-26 fixed-search batch. Counts describe only these coded rows. The fixed-search batch produced no row passing the target, split, evidence, metric, or consumer-boundary gate, so it supports claim control rather than an artifact-prevalence claim.

VII. ADMITTED EVIDENCE BUNDLE

Rows are admitted through a fixed path. In this paper, admitted means contract-admitted under the recorded workspace/runtime evidence level, not third-party full training-pipeline reproduction. A row must bind labeled samples to a stated target, expose metric provenance, attach finite-tail denominators, and carry consumer-boundary language. PIA, PIA-dropout, and DPDM W-1 have row-score replay in this workspace; recon and the main GSA row are admitted as source-documented point-estimate rows with their boundary language attached. This is why admitted rows can coexist with higher-scoring non-admitted candidates: admission is about reusable audit semantics, not metric magnitude.

Table V summarizes the admitted rows. The bundle is deliberately heterogeneous: black-box reconstruction and gray-box PIA baseline are primary leakage-risk evidence under their access modes, while stochastic dropout, GSA, and DPDM W-1 are admitted under comparator or bridge roles. Their shared unit is the evidence contract, not a common attack family or leaderboard rank. AUC uncertainty is reported only where the replay surface supports it: the sidecar gives bootstrap intervals for PIA baseline [0.8168, 0.8640], PIA-dropout [0.8039, 0.8535], and DPDM W-1 [0.4640, 0.5152].

Recon and the main GSA row remain point estimates, and H2 intervals remain candidate-side controls. The compact paper table lists caveats and a finite-tail basis; the machine-readable bundle retains quality/cost metadata for replay. We do not pool these rows into a single leakage estimate; each row supports only the consumer claim named by its boundary or caveat.

VIII. NON-ADMITTED H2 OUTPUT-CLOUD CANDIDATE

H2 is included as a stress test for claim admission: it shows that DiffAudit can preserve a technically strong signal while blocking the stronger consumer claim. The observable is deliberately narrower than a black-box attack claim: the target is queried repeatedly at fixed denoising strengths, and the scorer uses geometry among returned responses rather than query-to-output distance. The H2 response-strength cache contains repeated diffusion responses at timesteps 40, 80, 120, and 160. Instead of comparing a query image directly to outputs, the output-cloud scorer uses only geometry among outputs: within-step repeat RMSE, cross-timestep centroid RMSE, and small response-cloud PCA/Gram features. This design asks whether the response cloud itself carries membership information. The section also reports the negative portability gate: this is a candidate observable, not an admitted attack. Relative to

TABLE V

CONTRACT-ADMITTED DIFFAUDIT EVIDENCE ROWS. ROWS SHARE AN EVIDENCE CONTRACT, NOT A HOMOGENEOUS THREAT MODEL OR BENCHMARK LEADERBOARD. TAIL n_0 IS THE FINITE NONMEMBER BASIS FOR EMPIRICAL LOW-FPR READOUTS.

Track/role	Method	AUC	ASR	TPR@1%FPR	TPR@0.1%FPR	Tail n_0	Boundary/caveat
Black-box main	recon DDIM public-100 step30	0.837	0.740	0.220	0.110	$n_0 = 100$	runtime-mainline; public proxy risk evidence
Gray-box main	PIA GPU512 baseline	0.841	0.786	0.059	0.012	$n_0 = 512$	runtime-mainline; provenance caveat
Gray-box comp.	PIA + stochastic dropout	0.828	0.768	0.053	0.010	$n_0 = 512$	provisional defense comparator
White-box comp.	GSA 1k-3shadow	0.998	0.990	0.987	0.432	$n_0 = 1000$	upper-bound white-box comparator
White-box bridge	GSA against DPDM W-1	0.489	0.499	0.009	0.000	$n_0 = 1000$	runtime-smoke defense bridge

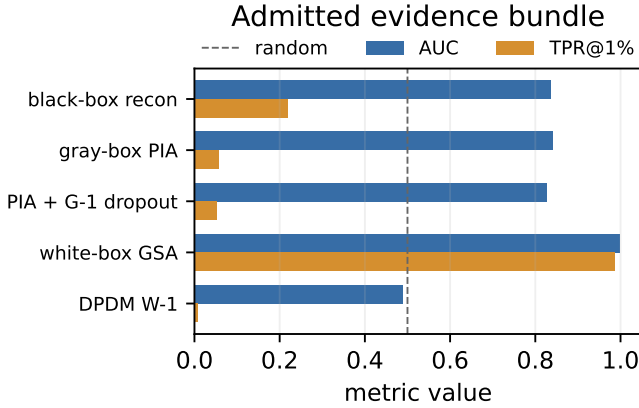


Fig. 3. AUC and TPR@1%FPR for admitted rows. The figure compares evidence under different access and defense contracts; it is not a single benchmark leaderboard.

black-box variation and reconstruction attacks that compare a query with returned or averaged outputs [15], [4], H2 removes query-to-output distance and scores output-output geometry. We therefore treat it as a response-contract mechanism probe, not a new black-box SOTA claim.

On the existing 512/512 cache, output-cloud logistic scoring reaches AUC 0.9615, ASR 0.9004, TPR@1%FPR 0.3340, and TPR@0.1%FPR 0.1172. It has higher point-estimate AUC than the tested raw H2 logistic baseline by 0.0558 and the tested lowpass H2 logistic baseline by 0.0659. A label shuffle returns random-level AUC 0.5076. Because the historical cache used class-ordered seed offsets, DiffAudit released only bounded 256/256 shared-position controls: with identical per-position seed offsets for members and nonmembers, the scorer still reaches AUC 0.9678 at seed 176 and 0.9562 at seed 177. A fold-disjoint same-family cross-cache transfer review gives AUC 0.9490 from seed 176 to 177 and AUC 0.9705 from seed 177 to 176. The feature families were fixed before metric replay, and the scorer does not use query-to-output distance. These controls do not exhaust every simple shortcut explanation or establish cross-model portability; they only support the bounded H2-family candidate claim.

The portability boundary is equally important. On the

SD/CelebA img2img 25/25 portability-test cache, output-cloud geometry reaches only AUC 0.7888 with zero strict-tail recovery ($n_0 = 25$) and underperforms the existing best simple-distance comparator. A smaller 10/10 stability cache is positive but still below simple distance. Thus the paper reports output-cloud geometry as a controlled mechanism candidate, not as an admitted black-box row: label shuffles collapse to random-level AUC as a sanity check, shared-position and cross-cache checks support only same-family robustness, and img2img blocks portability and product/runtime admission.

IX. NEGATIVE AND SUPPORT EVIDENCE

DiffAudit treats bounded non-admitted routes as measurement evidence when they answer a specific decision question. ReDiffuse STL-10 is a scoreable black-box reference point [15], but its denoising-loss and score-norm scouts are random-level. The local CommonCanvas response packet derived from the CopyMark/CommonCatalog route has a 50/50 contract [17], yet conditional denoising-loss reaches only AUC 0.5148. MIDST adds a tabular synthetic-data benchmark surface whose best available score remains below the reopen floor [18]. Tracing the Roots is positive (AUC 0.8158) [14], but it is a feature-packet replay, not raw image-level evidence. The collaborator Stable Diffusion ReDiffuse packet is replayable (AUC 0.7103), but source labels alone separate members and nonmembers perfectly, so it remains a cross-source stress test.

The decision value is route-specific. ReDiffuse STL-10 closes same-family expansion rather than refuting ReDiffuse; CommonCanvas shows that a second response contract alone is not portability; MIDST prevents cross-domain scoreability from becoming image-diffusion evidence; Tracing Roots remains feature-level support; the collaborator packet exposes source confounding; and the fixed-search batch records meta-data discovery without field-wide prevalence.

X. DISCUSSION AND LIMITATIONS

DiffAudit treats evidence contracts as measurement-validity conditions, not administrative metadata. For diffusion MIA, target identity, split semantics, row coverage, metric provenance, finite-tail denominator, and consumer boundary define what a score can mean as an audit claim. Without these controls, high AUC can come from source-label shortcuts,

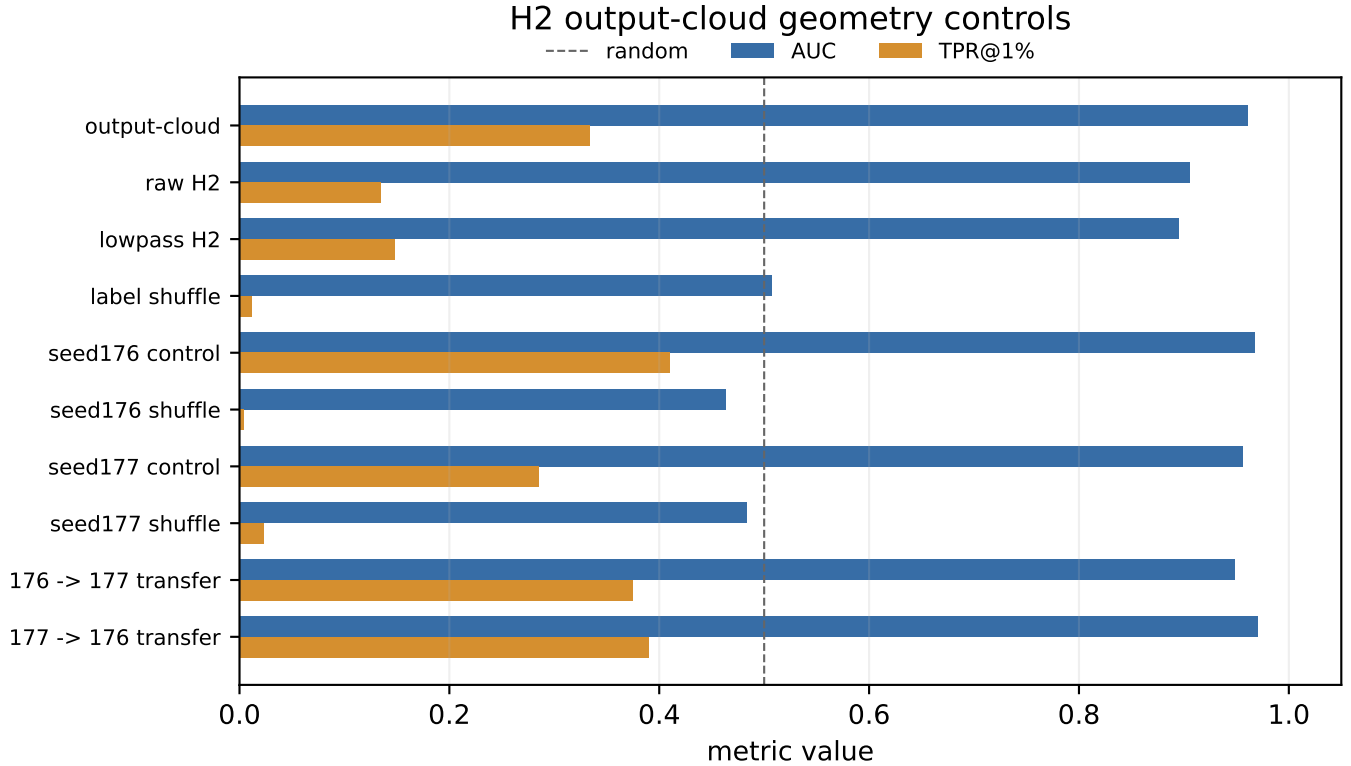


Fig. 4. H2 output-cloud geometry controls. The signal has higher point-estimate AUC than tested raw/lowpass baselines, collapses under label shuffle, and persists under seed-offset control, seed stability, and same-family cross-cache transfer. SD/CelebA img2img portability rows are summarized in text because they define the non-admission boundary.

prompt-conditioned auxiliary features, non-replayable scores, or finite-tail artifacts. The H2 case study shows that DiffAudit can still discover technical signals; the contract controls how far those signals may travel.

This paper does not claim broad generalization across all diffusion models. It does not claim that H2 output-cloud geometry is ready for downstream audit consumption. It does not claim that bounded weak scouts refute the original methods they resemble. The current evidence supports a case-study measurement result: H2 is a controlled candidate, fixed-search records are claim-control evidence, and neither supports portability, prevalence, or field-wide artifact claims without a separate frozen protocol.

A. Artifact Release Checklist

Reusable diffusion MIA artifacts should publish more than code and headline tables: target/checkpoint identity, immutable member and nonmember manifests, row-level scores or responses, metric code or JSON, nonmember denominators, hashes or commit IDs, and the intended consumer boundary. Trajectories, feature tensors, prompt sources, and regenerated samples should be named when they define the claim. This adapts reproducibility, datasheet, and model-card norms to row-bound privacy evidence [20], [21], [22].

B. Threats to Validity

Internal validity. Some results are produced by local replay scripts and repository artifacts rather than by re-running complete upstream training pipelines. The paper therefore claims score/response-level audit replay, not full upstream training-pipeline reproduction. DiffAudit mitigates this by recording row-level provenance and by refusing to call bounded scouts full reproductions or method refutations.

External validity. The admitted bundle is permission- and asset-bound. It should not be read as evidence that every diffusion model leaks at the same level, nor that one method dominates across modalities.

Metric validity. Low-FPR tails can be misleading on small packets. We therefore report them as finite empirical readouts and keep packet size and nonmember denominators attached to the claim.

Construct validity. DiffAudit measures whether a diffusion MIA result can be consumed as audit evidence under a stated contract. It does not measure all privacy harm, rank attack families globally, or prove training-data use for an individual datum. A high score may therefore remain non-admitted, and a weak bounded scout may still be scientifically useful if it closes a route-selection question.

Label and protocol validity. The gate labels are same-team adjudications, not independent inter-rater measurements.

We therefore use them for claim control and route decisions, not as prevalence estimates or field-quality rankings.

Publication bias. Negative routes were selected because they affected DiffAudit route decisions. The fixed-search batch and selected-corpus consistency pass reduce selection-history and label-coherence risk, but remain single-review, metadata-light traces. Standalone aggregate claims need a larger frozen corpus or a second independent label review.

XI. CONCLUSION

DiffAudit shows that diffusion MIA evidence is not exhausted by attack score. On the current evidence bank, the same contract admits five bounded contract rows, keeps a high-AUC H2 response-cloud signal candidate-only, and turns weak scouts, feature packets, fixed-search records, and source-confounded packets into claim-control evidence rather than overbroad conclusions. The reusable result is a reporting target for diffusion privacy work: publish the target, split, row-level observable, metric provenance, finite-tail basis, and consumer boundary, or state the narrower claim that the packet can actually support.

REFERENCES

- [1] J. Duan, F. Kong, S. Wang, X. Shi, and K. Xu, “Are diffusion models vulnerable to membership inference attacks?” in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023.
- [2] F. Kong, J. Duan, R. Ma, H. T. Shen, X. Shi, X. Zhu, and K. Xu, “An efficient membership inference attack for the diffusion model by proximal initialization,” in *The Twelfth International Conference on Learning Representations*, 2024.
- [3] S. Zhai, H. Chen, Y. Dong, J. Li, Q. Shen, Y. Gao, H. Su, and Y. Liu, “Membership inference on text-to-image diffusion models via conditional likelihood discrepancy,” in *Advances in Neural Information Processing Systems*, 2024.
- [4] Y. Pang and T. Wang, “Black-box membership inference attacks against fine-tuned diffusion models,” in *Network and Distributed System Security Symposium*, 2025.
- [5] Y. Pang, T. Wang, X. Kang, M. Huai, and Y. Zhang, “White-box membership inference attacks against diffusion models,” in *Proceedings on Privacy Enhancing Technologies*, vol. 2025, no. 2, 2025, pp. 398–415.
- [6] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, “Membership inference attacks against machine learning models,” in *2017 IEEE Symposium on Security and Privacy*, 2017, pp. 3–18.
- [7] S. Yeom, I. Giacomelli, M. Fredrikson, and S. Jha, “Privacy risk in machine learning: Analyzing the connection to overfitting,” in *2018 IEEE 31st Computer Security Foundations Symposium*, 2018, pp. 268–282.
- [8] N. Carlini, S. Chien, M. Nasr, S. Song, A. Terzis, and F. Tramèr, “Membership inference attacks from first principles,” in *2022 IEEE Symposium on Security and Privacy*, 2022, pp. 1897–1914.
- [9] J. Zhang, D. Das, G. Kamath, and F. Tramèr, “Position: Membership inference attacks cannot prove that a model was trained on your data,” in *2025 IEEE Conference on Secure and Trustworthy Machine Learning*, 2025, pp. 333–345.
- [10] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 6840–6851.
- [11] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, “High-resolution image synthesis with latent diffusion models,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 10 684–10 695.
- [12] N. Carlini, J. Hayes, M. Nasr, M. Jagielski, V. Schwag, F. Tramèr, B. Balle, D. Ippolito, and E. Wallace, “Extracting training data from diffusion models,” in *Proceedings of the 32nd USENIX Security Symposium*, 2023, pp. 5253–5270.
- [13] G. Somepalli, V. Singla, M. Goldblum, J. Geiping, and T. Goldstein, “Diffusion art or digital forgery? investigating data replication in diffusion models,” in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 6048–6058.
- [14] A. Floros, S.-M. Moosavi-Dezfooli, and P. L. Dragotti, “Tracing the roots: Leveraging temporal dynamics in diffusion trajectories for origin attribution,” 2025, arXiv:2411.07449.
- [15] J. Li, J. Dong, T. He, and J. Zhang, “Towards black-box membership inference attack for diffusion models,” in *Proceedings of the 42nd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 267. PMLR, 2025, pp. 34 271–34 287.
- [16] T. Dockhorn, T. Cao, A. Vahdat, and K. Kreis, “Differentially private diffusion models,” *Transactions on Machine Learning Research*, 2023.
- [17] C. Liang and J. You, “Real-world benchmarks make membership inference attacks fail on diffusion models,” 2024, arXiv:2410.03640.
- [18] M. Shafieinejad, X. He, M. Alinoori, J. Jewell, S. Ayromlou, W. Pang, V. Chatrath, G. Sharma, and D. Pandya, “MIDST challenge at SaTML 2025: Membership inference over diffusion-models-based synthetic tabular data,” 2026, arXiv:2603.19185.
- [19] J. Dubiński, A. Kowalczyk, F. Boenisch, and A. Dziedzic, “CDI: Copyrighted data identification in diffusion models,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 18 674–18 684.
- [20] J. Pineau, P. Vincent-Lamarre, K. Sinha, V. Larivière, A. Beygelzimer, F. d’Alché Buc, E. Fox, and H. Larochelle, “Improving reproducibility in machine learning research (a report from the neurips 2019 reproducibility program),” *Journal of Machine Learning Research*, vol. 22, no. 164, pp. 1–20, 2021.
- [21] T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. Daumé III, and K. Crawford, “Datasheets for datasets,” *Communications of the ACM*, vol. 64, no. 12, pp. 86–92, 2021.
- [22] M. Mitchell, S. Wu, A. Zaldivar, P. Barnes, L. Vasserman, B. Hutchinson, E. Spitzer, I. D. Raji, and T. Gebru, “Model cards for model reporting,” in *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 2019, pp. 220–229.